

Chapter 1

Basic Discrete Structure

Sets: The theory of set was originated in 1895 by the German mathematician **G.Cantor** who defined set as a collection of definite and distinguishable objects selected by means of some rules or description. The theory of sets is the basis for all the branches of modern mathematics.

We use capital letters **A,B,C.....X,Y,Z** to denote sets and small letters **a,b,c.....x,y,z** to denote elements of sets. If an element 'a' is a member of set A then we write $a \in A$.

Representation of Sets: A set may be represented by the following methods

1. **Description Method:** In this method a set is specified by a verbal description. For example, the set S of numbers 1,2,3 is designated as
 $S = \text{"The set of positive integers less than 4"}$
2. **Tabulation Method:** In this method, a set is specified by listing all elements in a set. Thus, we can write the set S of numbers 1,2 and 3 as:
 $S = \{1,2,3\}$.
3. **Rule Method or Set-Builder Method:** In this method a set is specified by stating a characteristic property common to all elements in the set. The above set can be represented as:
 $S = \{x: x \text{ is an integers and } 1 \leq x \leq 3\}$

Definitions and sets Terminologies

Finite Set: A set consisting of finite number of elements is called finite set. Thus, the set of days in a week is a finite set.

Infinite Set: A set consisting of infinite number of elements is called infinite set. A set of all odd numbers is called an infinite set.

Empty Set: A set without any element is called an empty set or null set or void set and is usually denoted by the symbol ϕ .

$P = \{x: x \text{ is the male students of PK campus}\} = \phi$

Unit Set: A set consisting of only one element is called a unit set or singleton set. Example of unit set is:

$S = \{2\}$. A set of even primes

Universal Set: A universal set is the original set that contains all elements under consideration of a particular situation and is denoted by U .

Subset: If A and B are sets such that every element of A is also an element of B then A is said to be a subset of B and is denoted by $A \subseteq B$.

Disjoint Set: Two sets are called disjoint set if their intersection is the empty set.

Example: Let $A = \{1,3,5,7,9\}$ and $B = \{2,4,6,8,10\}$. Because $A \cap B = \phi$, A and B are disjoint.

Power Set: If S is any set then the family of all the subsets of S is called the power set of S . The power set of S is denoted by $P(S)$.

Obviously, ϕ and S are both elements of $P(S)$. If the set S is finite and contain n elements then the power set of S will then contain 2^n elements

Example: What is the power set of the set $\{0,1, 2\}$.

Solution: The power set $P(\{0,1,2\})$ is the set of all subsets of $\{0,1,2\}$. Hence $P(\{0,1,2\}) = \{\phi, \{0\}, \{1\}, \{2\}, \{0,1\}, \{0,2\}, \{1,2\}, \{0,1,2\}\}$.

Example: What is the power set of the empty set? What is the power set of the set $\{\phi\}$?

Solution: The empty set has exactly one subset, namely itself. Consequently, $P(\phi) = \{\phi\}$.

The set $\{\phi\}$ has exactly two subsets, namely, ϕ and the set $\{\phi\}$ itself. Therefore,

$$P(\{\phi\}) = \{ \phi, \{ \phi \} \}$$

If a set has n elements then its power set has 2^n elements.

Ordered Pair: A set of two elements 'a' and 'b' written as $\{a,b\}$ is called a pair. If we consider the order of the elements, the pair is called ordered pair and denoted as (a,b) . Here 'a' is called the first element and 'b' is called the second element of the ordered pair (a,b) .

In ordered pair $(a,b) \neq (b,a)$

Cartesian product of Two Sets

The Cartesian product of two non-empty sets A and B is defined as the set of all possible ordered pairs (a,b) such that $a \in A$ and $b \in B$

Mathematically, we can write

$$A \times B = \{(a,b) : a \in A, b \in B\}$$

In general, $A \times B \neq B \times A$

Example: Let $A = \{1,2\}$ and $B = \{3,4,5\}$. Construct $A \times B$ and $B \times A$.

Solution:

$$A \times B = \{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5)\}$$

$$B \times A = \{(3,1), (3,2), (4,1), (4,2), (5,1), (5,2)\}$$

Venn diagram: Sets, subsets and set operations can be represented by diagrams called Venn diagram. In Venn diagram, we usually represent the universal set by a rectangle and its subsets by circles. The elements of U are represented by points within the rectangle while the elements of the subsets of U are represented by the points within the circle.

Set Operations: Sets may be combined and operated in various ways to form new sets. The basic operations on set are

1. **Union:** The union of two sets A and B , denoted by $A \cup B$, is the set of only those elements which belongs to either A or B or both A and B . Symbolically we write it as:
 $A \cup B = \{x: x \in A \text{ or } x \in B \text{ or } x \in A \text{ and } B\}$.
For example, If $A = \{1,3\}$ and $B = \{3,4,5\}$ then $A \cup B = \{1,3,4,5\}$
2. **Intersection:** The intersection of two sets A and B , denoted as $A \cap B$, is the set of only those elements which belongs to both A and B . In symbol, we write
 $A \cap B = \{x: x \in A \text{ and } x \in B\}$
For example, if $A = \{2, 3, 4\}$ and $B = \{3,5\}$ then $A \cap B = \{3\}$
3. **Complement:** Let A be the subset of a universal set U . Then the complement of A with respect to U is the set of all those elements of U which do not belong to A and is denoted by A' or A^c . In symbols, we write $A^c = \{x: x \in U \text{ and } x \notin A\}$
For example, if $U = \{1, 2, 3, 4, 5\}$ and $A = \{2, 4\}$ then $A' = \{1, 3, 5\}$
4. **Difference:** Let A and B be two sets and each set is the subset of a universal set U . Then, A difference B denoted by $A - B$ is the set of all those elements which belong to A but not B . In general, we write this as:
 $A - B = \{x: x \in A \text{ and } x \notin B\}$
Example: If $A = \{a, b, x, y\}$ and $B = \{c, d, x, y\}$ then
 $A - B = \{a, b\}$ and $B - A = \{c, d\}$
Thus $A - B \neq B - A$

Set Identities

- I. Laws Governing Union and Intersection
 1. Identity Laws

- a. $A \cup \phi = A$
- b. $A \cup U = U$
- c. $A \cap \phi = \phi$
- d. $A \cap U = A$

2. Idempotent Laws

- a. $A \cup A = A$
- b. $A \cap A = A$

3. Commutative Laws

- a. $A \cup B = B \cup A$
- b. $A \cap B = B \cap A$

4. Associative Laws

- a. $A \cup (B \cap C) = (A \cup B) \cap C$
- b. $A \cap (B \cup C) = (A \cap B) \cup C$

5. Distributive Laws

- a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

II. Laws Governing complements

- a. $(A')' = A$
- b. $A \cup A' = U$
- c. $A \cap A' = \phi$
- d. $\phi' = U$
- e. $U' = \phi$

III. Laws Governing set Difference

- 1. $A - B = A \cap B'$
- 2. $U - A = A'$
- 3. $A - U = \phi$
- 4. $A - \phi = A$
- 5. $\phi - A = \phi$
- 6. $A - A = \phi$

IV. De Morgan's Laws

- a. $(A \cup B)' = A' \cap B'$
- b. $(A \cap B)' = A' \cup B'$

Example: Given $A = \{1,3,5\}$, $B = \{0,1,2,3\}$ and $C = \{0,1,5\}$. Verify the distribution laws.

Solution:

The distributive laws for union and intersection are:

- a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$b. A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$a) \text{ We have } A \cup B = \{0, 1, 2, 3, 5\}, A \cap B = \{1, 3\}$$

$$A \cup C = \{0, 1, 3, 5\}, A \cap C = \{0, 1\}$$

$$B \cup C = \{0, 1, 2, 3, 5\}, B \cap C = \{0, 1\}$$

$$L S = A \cup (B \cap C) = \{1, 2, 3\} \cup \{0, 1\} = \{0, 1, 3, 5\}$$

$$R S = (A \cup B) \cap (A \cup C) = \{0, 1, 2, 3, 5\} \cap \{0, 1, 3, 5\} = \{0, 1, 3, 5\}$$

$$b) L S = A \cap (B \cup C) = \{1, 3, 5\} \cap \{0, 1, 2, 3, 5\} = \{1, 3, 5\}$$

$$R S = (A \cap B) \cup (A \cap C) = \{1, 3\} \cup \{1, 5\} = \{1, 3, 5\}$$

Example 2: Prove that $(A \cap B)' = A' \cup B'$ using set builder notation

$$\text{Solution: } (A \cap B)' = \{x: x \in (A \cap B)'\}$$

$$= \{x: x \notin (A \cap B)\}$$

$$= \{x: \neg(x \in A \cap B)\}$$

$$= \{x: \neg(x \in A \wedge x \in B)\}$$

$$= \{x: \neg(x \in A) \vee \neg(x \in B)\}$$

$$= \{x: x \notin A \vee x \notin B\}$$

$$= \{x: x \in A' \vee x \in B'\}$$

$$= \{x: x \in A' \cup B'\}$$

$$= A' \cup B'$$

Example: Prove that $(A \cup B)' = A' \cap B'$

Solution:

Inclusion and Exclusion and Application

The inclusion-exclusion principle in set theory is a counting technique which generalizes the method of obtaining the number of elements in the union of two or more finite sets; symbolically expressed as:

$|A \cup B| = |A| + |B| - |A \cap B|$ where A and B are finite sets and $|A|$ denotes number of elements in A.

The principle is more clearly seen in case of three sets.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Example 1: There are 345 students at a college who have taken a course in calculus, 212 who have taken a course in discrete mathematics, and 188 who have taken course in both calculus and discrete mathematics. How many students have taken the course in either calculus or discrete mathematics?

Solution: Here we have number of students taking a course in calculus $|C| = 345$

Number of students taking a course in discrete mathematics $|D| = 212$

Number of students taking course in both calculus and discrete mathematics $|C \cap D| = 188$

Number of students taking either discrete mathematics or calculus

$$|C \cup D| = |C| + |D| - |C \cap D|$$

$$\text{Or } |C \cup D| = 345 + 212 - 188 = 369$$

Example 2:

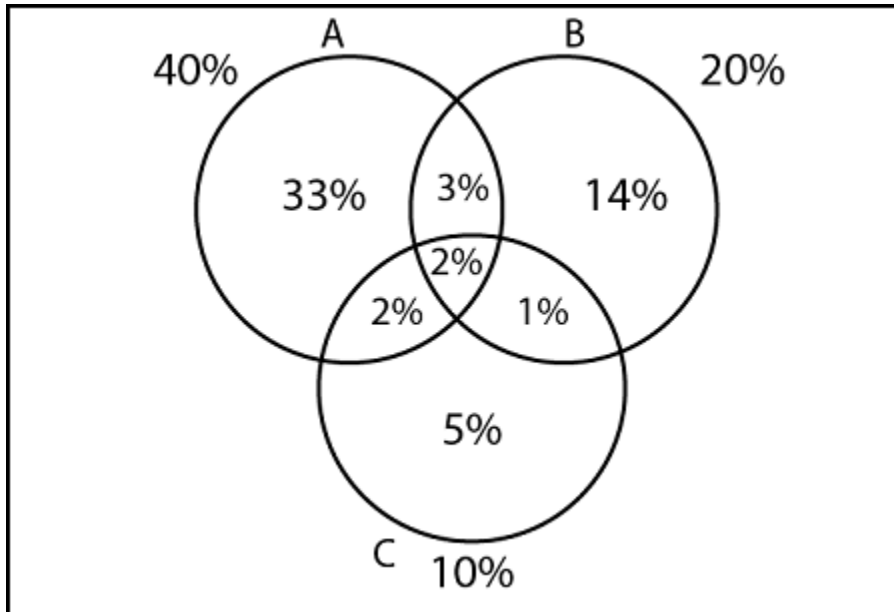
In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the number of families which buy

1. Number of families which buy all three newspapers.
2. Number of families which buy newspaper A only
3. Number of families which buy newspaper B only
4. Number of families which buy newspaper C only
5. Number of families which buy None of A, B, C
6. Number of families which buy exactly only one newspaper
7. Number of families which buy newspaper A and B only
8. Number of families which buy newspaper B and C only
9. Number of families which buy newspaper C and A only
10. Number of families which buy at least two newspapers

11. Number of families which buy at most two newspapers

12. Number of families which buy exactly two newspapers

Solution:



1. Number of families which buy all three newspapers:

1. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

2. $n(A \cup B \cup C) = 40 + 20 + 10 - 5 - 3 - 4 + 2 = 60\%$

2. Number of families which buy newspaper A only

1. $= 40 - 7 = 33\%$

3. Number of families which buy newspaper B only

1. $= 20 - 6 = 14\%$

4. Number of families which buy newspaper C only

$$1. = 10 - 5 = 5\%$$

5. Number of families which buy None of A, B, and C

$$n(A \cup B \cup C)^c = 100 - n(A \cup B \cup C)$$

$$n(A \cup B \cup C)^c = 100 - [40 + 20 + 10 - 5 - 3 - 4 + 2]$$

$$n(A \cup B \cup C)^c = 100 - 60 = 40\%$$

6. Number of families which buy exactly only one newspaper

$$1. = 33 + 14 + 5 = 52\%$$

7. Number of families which buy newspaper A and B only

$$1. = 3\%$$

8. Number of families which buy newspaper B and C only

$$1. = 1\%$$

9. Number of families which buy newspaper C and A only

$$1. = 2\%$$

10. Number of families which buy at least two newspapers

$$1. = 8\%$$

11. Number of families which buy at most two newspapers

$$1. = 98\%$$

12. Number of families which buy exactly two newspapers

$$1. = 6\%$$

Computer Representation of Sets

One method to represent sets using computer is to store the elements of the set in an unordered fashion. In this method, the operations like union, intersection, or difference of two sets would be time consuming since large amount of time will be required for searching of elements. A method using an arbitrary ordering of the elements of the universal set to store the elements can overcome this problem.

When a universal set U is finite say $U = \{x_1, x_2, x_3, \dots, x_n\}$ then all subset U can be represented with the bit string of length n . A bit string is a string over the alphabet $\{0,1\}$. If A is a subset of U , then A is represented by bit string of length n where i^{th} bit in this string is 1 if x_i belongs to A and is 0 if x_i does not belong to A .

Example: Let $U = \{1,2,3,4,5,6,7,8,9,10\}$

- i. What bit strings represent the subset of all odd integers in U ?

Solution: The set of odd integers is $S = \{1, 3, 5, 7, 9\}$. The bit string that represents the set S has a bit 1 in first, third, fifth, seventh and ninth position. It is 1 0 1 0 1 0 1 0 1 0.

- ii. What bit strings represent the subset of all even integers in U ?

Solution: The set of even integers is $E = \{2, 4, 6, 8, 10\}$. The bit string that represents the set E has a bit 1 in second, fourth, sixth, eighth and tenth position. It is 0 1 0 1 0 1 0 1 0 1.

Example : The bit strings for the sets $\{1,2,3,4,5\}$ and $\{1,3,5,7,9\}$ are 1 1 1 1 1 0 0 0 0 0 and 1 0 1 0 1 0 1 0 1 0. Find the union and intersection of sets using given bit strings

Solution:

Union: The bit string for the union of these two sets is: 1 1 1 1 1 0 0 0 0 0 v 1 0 1 0 1 0 1 0 1 0

= 1 1 1 1 1 0 1 0 1 0 which corresponds to the set $\{1,2,3,4,5,7,9\}$

Intersection: The bit string for the intersection of these two sets is:

1 1 1 1 1 0 0 0 0 0 ^ 1 0 1 0 1 0 1 0 1 0 = 1 0 1 0 1 0 0 0 0 0 which corresponds to the set $\{1,3,5\}$.

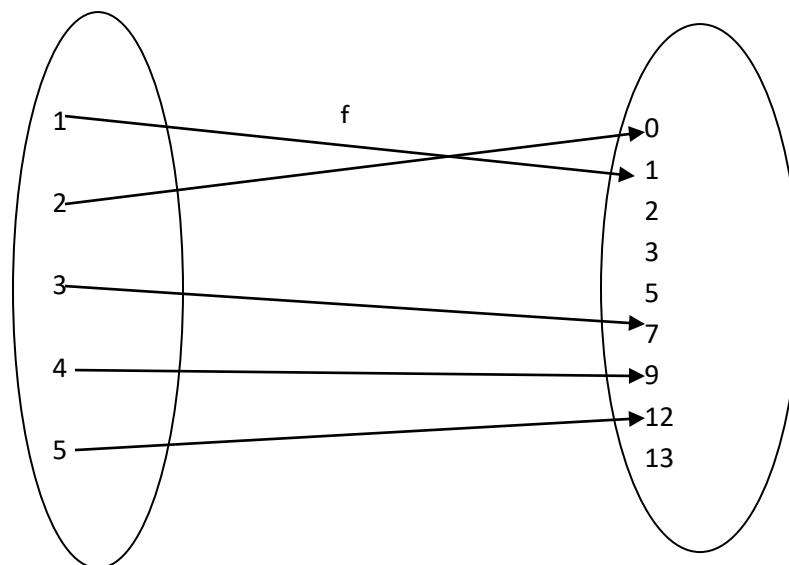
1.2 Function

Basic Concepts

Let A and B be two non-empty sets. A function f from A to B is a set of ordered pairs with the property that for each element x in A there exists a unique element y in B . The set A is called the domain of the function and the set B is called co-domain. If $f(x, y) \in f$ then it is customary to write $y = f(x)$, y is called the image of x and x is called pre-image of y . The set consisting of all the images of the elements of set A under function f is called the range of f . It is denoted by $f(A)$.

The concept of function is extremely used in mathematics and in computer science. For instance functions are used to represent how long it takes to solve a problem of a given size. Many computer programs and subroutines are designed to calculate the values of functions.

Example: Let $A = \{1, 2, 3, 4, 5\}$, $B = \{0, 1, 2, 3, 5, 7, 9, 12, 13\}$ and $f = \{(1, 1), (2, 0), (3, 7), (4, 9), (5, 12)\}$ then f is a function from A to B because each element of A has a unique image in B which can be expressed by a diagram as:



Sum and Product of Functions

Two real-valued functions with the same domain can be added and multiplied.

Definition: Let f_1 and f_2 be functions from A to R . Then f_1+f_2 and $f_1.f_2$ are also functions from A to R defined by

$$(f_1+f_2)(x) = f_1(x)+f_2(x).$$

$$(f_1f_2)(x) = f_1(x).f_2(x)$$

The functions f_1+f_2 and f_1f_2 have been defined by specifying their values at x in terms of the values of f_1 and f_2 at x .

Example: Let f_1 and f_2 be functions from R to R such that $f_1(x) = x^2$ and $f_2(x) = x-x^2$. What are the functions f_1+f_2 and f_1f_2 ?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1+f_2)(x) = f_1(x)+f_2(x) = x^2+(x-x^2) = x.$$

And

$$(f_1f_2)(x) = x^2(x-x^2) = x^3-x^4$$

Types of Function

Injective Function (One to one):- A function from set A to set B is one to one if for all $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$ implies $x_1=x_2$.

Example: Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c)=1$, and $f(d) = 3$ is one to one.

Solution: The function is one to one because f takes on different values at the four elements of its domain.

Example: Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one to one or not?

Solution: The function $f(x) = x^2$ is not one to one because for instance $f(1) = 1$ and $f(-1)=1$ but $1 \neq -1$.

Example: Determine whether the function $f(x) = x+1$ from the set of integers to itself is one to one or not

Solution: The function $f(x) = x+1$ is one to one function. To explain this note that $x+1 \neq y+1 \rightarrow x \neq y$

Surjective Function (On to Function):- A function from set A to set B is onto if every element of B is the image of some element in A .

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1,2,3\}$ defined by $f(a) = 3, f(b) = 3, f(c) = 1$ and $f(d) = 3$. Is f an onto function?

Solution: Because all three members of the co-domain are images of elements in the domain, we see that f is onto. If the co-domain were $\{1,2,3,4\}$ then f would not be onto.

Example: Is the function $f(x) = x^2$ from the set of integers to set of integers onto?

Solution: The function is not onto because for every integer y there is no integer x with $x^2 = -1$ for instance.

Example: Is the function $f(x) = x+1$ from the set of integers to the set of integers onto?

Solution: This function is onto because for every integer y there is an integer x such that $f(x) = y$. To see this, note that $f(x) = y$ if and only if $x+1 = y$ which holds if and only if $x = y-1$.

Bijjective Function (One to one correspondence): A function from set A to set B is called one to one correspondence if it is both injective and surjective

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1,2,3,4\}$ with $f(a) = 4, f(b) = 2, f(c) = 1$ and $f(d) = 3$. Is f a bijection?

Solution: The function is one to one and onto. It is one to one because no two values in the domain are assigned the same function value. It is onto because all four elements of its co-domain are images of elements in the domain. Hence, f is a bijection

Inverse Functions

Consider a one to one correspondence f from the set A to the set B . Because f is an onto function, every element of B is the image of some element in A . Furthermore, because f is also a one to one function, every element of B is the image of a unique element of A . Consequently, we can define a new function from B to A that reverses the correspondence given by f .

Definition: Let f be a one to one correspondence from the set A to the set B . The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$

If a function is not a one to one correspondence, we cannot define an inverse function of f . When f is not a one to one correspondence, either it is not one to one or it is not onto, we cannot define inverse function of f . If f is not one to one, some element b in the co-domain is the image of more than one element in the domain. If f is not onto, for some element b in the co-domain,

no element a in the domain exists for which $f(a) = b$. Consequently, if f is not a one to one correspondence, we cannot assign to each element b in the co-domain a unique element a in the domain such that $f(a) = b$.

A one to one correspondence is called invertible because we can define an inverse of this function. A function is not invertible if it is not a one to one correspondence because the inverse of such a function does not exist.

Example : Let f be the function from $\{a,b,c\}$ to $\{1,2,3\}$ such that $f(a) = 2$, $f(b) = 2$ and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is one to one correspondence. The inverse function f^{-1} reverses the correspondences given by f . So, $f^{-1}(1) = c$, $f^{-1}(2) = a$ and $f^{-1}(3) = b$.

Example: Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be such that $f(x) = x+1$. Is f invertible and if it is, what is its inverse?

Solution: The function f has inverse it is one to one correspondence as it is both one to one and on to. To reverse the correspondence, suppose that y is the image of x , so that $y = x+1$. Then $x = y-1$. This means that $y-1$ is the unique element of $\mathbb{Z}$ that is sent to y by f . Consequently, $f^{-1}(y) = y-1$.

Example: Let f be the function from $\mathbb{R}$ to $\mathbb{R}$ with $f(x) = x^2$. Is f invertible?

Solution: Because $f(-2) = f(2) = 4$, f is not one to one. If an inverse of a function were defined, it would have to assign two elements to 4. Hence, f is not invertible.

Example: Let f be the function from the set of all non-negative real numbers to the set of all non-negative real numbers with $f(x) = x^2$. Is f invertible?

Solution: The function $f(x) = x^2$ from set of non-negative real numbers to the set of non-negative real numbers is one to one. To see that, note that $f(x) = f(y)$ then $x^2 = y^2 \rightarrow (x+y)(x-y)=0$.

This means that $x+y = 0$ or $x-y=0$.

Or $x=-y$ or $x=y$

Because both x and y are non-negative, we must have $x=y$. So this function is one to one.

Furthermore, $f(x) = x^2$ is onto when the co-domain is the set of all non-negative real numbers because each non-negative real number has a square root. That is, if y is a non-negative real number, there exists a non-negative real number s such that $x = \sqrt{y}$, which means that $x^2=y$.

Because the function $f(x) = x^2$ from the set of non-negative real numbers to the set of non-negative real numbers is one to one and onto, it is invertible. Its inverse is given by the rule $f^{-1}(y) = \sqrt{y}$.

Example: Show that the mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = ax+b$ where $a, b, c \in \mathbb{R}$, $a \neq 0$, is invertible. Define its inverse.

Solution: The inverse of a function exists if that function is both one-one and onto. Therefore, we first show f is one to one and we show it is onto.

For, if $x_1, x_2 \in \mathbb{R}$ then $f(x_1) = f(x_2)$

$$\rightarrow ax_1+b = ax_2+b$$

$$\rightarrow x_1=x_2$$

This proves f is one-to one.

Again, if $y \in \mathbb{R}$, $y = f(x) = ax+b$

$$x = (y-b)/a \in \mathbb{R}$$

$$\text{and } f(y-b)/a = a.(y-b)/a+b = y$$

Therefore y is the image of $(y-b)/a$.

Thus, f is onto.

Hence f is one to one and onto therefore f^{-1} exists and is defined by $f^{-1}(y) = (y-b)/a$. Here y is just a dummy variable and can be replaced by x . Then we have $f^{-1}(x) = (x-b)/a$

Example: Let: $\mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x-7$. Find a formula for the inverse function if it is invertible.

Solution: Suppose that $x_1, x_2 \in \mathbb{R}$ and $f(x_1) = f(x_2)$

$$\text{Then } f(x_1) = f(x_2) \text{ implies } 3x_1-7 = 3x_2-7$$

$$\text{Implies } x_1 = x_2.$$

Hence $f(x)$ is one to one.

$$\text{Also let } y \in \mathbb{R} \text{ then } y = f(x) = 3x-7$$

$$\rightarrow x = (y+7)/3$$

$$\text{Now } f(y+7)/3 = 3(y+7)/3-7 = y$$

Thus f is onto.

Since f is both one to one and onto. So f^{-1} exists and is defined by $f^{-1}(y) = y+7/3$

In terms of x , the inverse function is $f^{-1}(x) = (x+7)/3$.

Composition of Function: Let g be a function from the set A to B and f be a function from the set B to the set C . The composition of the functions f and g , denoted by $f \circ g$, is defined by $(f \circ g)(a) = f(g(a))$.

Example: Let f and g be two functions from the set of integers to set of integers defined by $f(x) = 2x+3$ and $g(x) = 3x+2$. What is the composition of f and g ? What is the composition of g and f ?

Solution:

Composition of f and g is defined as

$$f \circ g = f(g(x)) = f(3x+2) = 2(3x+2)+3 = 6x+7$$

And

$$g \circ f = g(f(x)) = g(2x+3) = 3(2x+3) = 6x+11$$

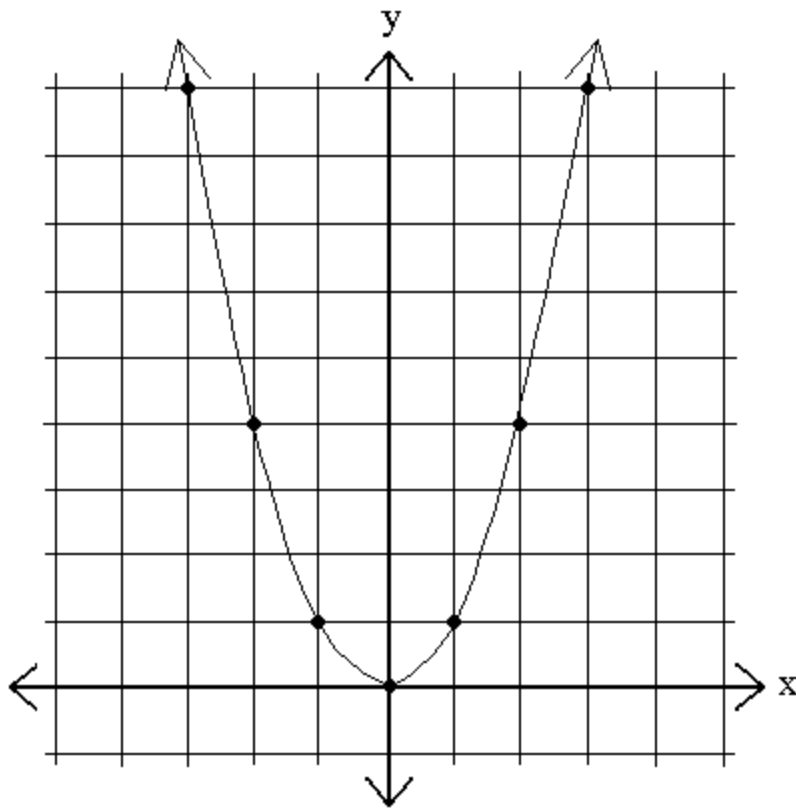
The Graphs of Functions

We can associate a set of pairs in $A \times B$ to each function from A to B . This set of pairs is called the graph of the function and is often displayed pictorially to aid in understanding the behavior of the function.

Definition: Let f be a function from the set A to the set B . The graph of the function f is the set of ordered pairs $\{(a,b) | a \in A \text{ and } f(a) = b\}$

Example: Display the graph of the function $f(x) = x^2$ from the set of integers to the set of integers.

Solution: The graph of f is the set of ordered pairs of the form $(x, f(x)) = (x, x^2)$ where x is an integer. This graph is displayed in figure below.



Functions for Computer Science

Floor Function

Let x be a real number less than $\lfloor x \rfloor$ is called floor functions of x , assigns to the real number x the largest integer that is less than or equal to x . This function rounds x down to the closest integer less than or equal to x .

Ceiling Function: If x is a real number then $\lceil x \rceil$ called the ceiling function of x assigns to the real number x the smallest integer that is greater than or equal to x . This function rounds x up to the closest integer greater than or equal to x .

Example: Data stored on a computer disk are transmitted over a data network are usually represented as string of bytes. Each byte is made up of 8 bits. How many bytes are required to encode 300 bits of data?

Solution: To determine the number of bytes needed, we determine the smallest integer that is at least as large as the quotient when 300 is divided by 8. Therefore, the number of bytes required $\lceil 300/8 \rceil = \lceil 37.5 \rceil = 38$

Example: In ATM, data are organized into cells of 55 bytes. How many ATM cells can be transmitted in one minute over a connection that transmits data at the rate of 600 kilobits per second?

Solution: In 1 minute, this connection can transmit $600000 \times 60 = 36000000$ bits. Each ATM cell is 55 bytes long which means that it is $55 \times 8 = 440$ bits long. To determine the number of cells that can be transmitted in 1 minute, we determine the largest integer not exceeding the quotient when 36000000 is divided by 440. Consequently, $\lfloor 36000000/440 \rfloor = 81818$ ATM cell can be transmitted in 1 minute over 600,000 kilobit per second connection.

Boolean Function

Boolean algebra deals with binary variables and logic operation. A **Boolean Function** is described by an algebraic expression called **Boolean expression** which consists of binary variables, the constants 0 and 1, and the logic operation symbols. Consider the following example.

$$\begin{array}{lll} F(A, B, C, D) & = & A + \overline{BC} + ADC \\ \text{Boolean Function} & & \text{Boolean Expression} \end{array} \quad \text{Equation No. 1}$$

Here the left side of the equation represents the output Y. So we can state equation no. 1

$$Y = A + \overline{BC} + ADC$$

Exponential Function

If b is any number such that $b > 0$ and $b \neq 1$ then an exponential function is a function in the form $f(x) = b^x$.

Where b is called the base and x can be any real number.

The property of exponential function is that it grows very fast as there is small increase in the value of variable.

Fuzzy Set

Fuzzy sets can be considered as an extension and gross oversimplification of classical sets. It can be best understood in the context of membership. Basically it allows partial membership which means that it contain elements that have varying degrees of membership in the set. From this, we can understand the difference between classical set and fuzzy set. Classical set theory allows the membership of the elements in the set in binary terms

Fuzzy set theory permits membership function in the interval $[0,1]$.

Fuzzy set is defined as follows. If X is an universe of discourse and x is a particular element of X then a fuzzy set A defined on X and can be written as a collection of ordered pair $A = \{(x, \mu_A(x), x \in X)\}$

Membership Function

For any set X , a membership function on X is any function from X to the real unit interval $[0,1]$. The membership function which represents a fuzzy set A is denoted by μ_A . For an element x of set X , the value $\mu_A(x)$ is called the membership degree of x in the fuzzy set A . The membership degree $\mu_A(x)$ quantifies the grade of membership of the element x to the fuzzy set A .

The value 0 means that x is not a member of the fuzzy set, the value 1 means that x is a fully a member of the fuzzy set. The values between 0 and 1 characterize fuzzy members which belong to fuzzy set only partially.

Fuzzy Set Operation

Intersection: The membership function of the intersection of two fuzzy sets A and B with membership functions μ_A and μ_B respectively is defined as the minimum of the two individual membership functions. This is called the minimum criterion.

$$\mu_{A \cap B} = \min(\mu_A, \mu_B)$$

For example, in binary logic, union represents the logical AND operation with the truth table:

X	Y	X AND y
0	0	0
0	1	0
1	0	0
1	1	1

But in case of *fuzzy* set with the given membership functions, the intersection is computed as follows using above definitions:

X	Y	min(x,y)
0	0	0
0	1	0
1	0	0
1	1	1
0.2	0.5	0.2

0.7	0.2	0.2
0.6	0.6	0.6

Unions

The membership function of the union of two fuzzy sets A and B with membership functions μ_A and μ_B respectively is defined as the maximum of two individual membership functions. This is called the maximum criterion.

$$\mu_{A \cup B} = \max(\mu_A, \mu_B)$$

For example: In binary logic, union represents the logic OR operation with the truth table as follows:

X	Y	X OR y
0	0	0
0	1	1
1	0	1
1	1	1

But in case of fuzzy set with the given membership functions, the union is computed as follows using above definition:

X	Y	max(x,y)
0	0	0
0	1	1
1	0	1
1	1	1
0.2	0.5	0.5
0.7	0.2	0.7
0.6	0.6	0.6

Complement: The membership function of the complement of a Fuzzy set with membership function μ_A is defined as the negation of the specified membership function. This is called the negation criterion.

$$\mu_{A'} = 1 - \mu_A$$

X	NOT x
---	-------

For	0	1
	0	1
with	1	0
	1	0

example, in binary logic, union represents the logical OR operation the following truth table:

But in case of fuzzy set with the given membership functions, the complement is computed as follows:

x	1-x
0	1
1	0
0.2	0.8
0.7	0.3
0.6	0.4

Sequences and Summations

Introduction: Sequences are ordered lists of elements. Sequences are used in discrete mathematics in many ways such as to represent solutions to counting problems. They are also important data structure in computer science. Here we review the notation used to represent sequences and sums of terms of sequences

A sequence is a function from a subset of the set of integers to a set **S**. We use the notation a_n to denote the image of the integer n . We call a_n a term of the sequence

We use the notation $\{a_n\}$ to describe the sequence. Here a_n represents an individual terms of the sequence $\{a_n\}$. Here choice of letter is arbitrary.

Example: Consider the sequence $\{a_n\}$, where $a_n = \frac{1}{n}$

The list of the terms of this sequence, beginning with a_1 , namely,

$a_1, a_2, a_3, a_4, \dots$

starts with

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

Geometric Progression: A geometric progression is a sequence of the form $a, ar, ar^2, ar^3, \dots, ar^n, \dots$

where the initial term a and the common ratio r are real numbers.

Example: The sequence $\{a_n\}$ with $a_n = 2^n$ is a geometric progression. The list of terms are : $b_1 = 2, b_2 = 4, b_3 = 8$ and so

Arithmetic Progression: An arithmetic progression is a sequence of the form

$A, a+d, a+2d, a+3d, \dots, a+nd$

where the initial term a and the common difference d are real numbers.

Example: The sequences $\{s_n\}$ with $s_n = -1+4n$ is an arithmetic progressions with initial terms and common difference equal to 3 and 4 respectively if we start with $n=0$.

Sequences of the form $a_1, a_2, a_3, \dots, a_n$ are often used in computer science. These finite sequences are also called strings. This string is also denoted as $a_1 a_2 a_3 \dots a_n$. The length of string S is the number of terms. The empty string has length zero. The string "abcd" is a string of length four.

Special Integer Sequences

A common problem in discrete mathematics is finding a formula or a general rule for constructing the terms of a sequence. Sometimes only a few terms of a sequence solving a problem are known, the goal is to identify the sequence. Even though the initial terms of a sequence do not determine the entire sequence, knowing the first few terms may help to make an educated conjecture about the identity of your sequence.

Example: Find formulae for the sequence with the following five terms : $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{16}, \dots$

Solution: The denominators are power of 2. The sequence with $a_n = a/2^n, n=0,1,2,3,4,5, \dots$ is a possible math. The proposed sequence is a geometric progression with $a=1$ and $r=1/2$.

Example: How can we produce the terms of the sequence if the first ten terms are $5, 11, 17, 23, 29, 35, 41, 47, 53, 59$?

Solution: Here each higher term in the sequence is 6 more than its previous term. Hence the n th term is given by formula $a_n = 5+6(n-1)$

Example: Write down the sequence of even numbers.

Solution: An integer is even if it is divisible by two. For example, 2 is even because the result of dividing it by itself is 1.

A formal definition of an even number is that it is an integer of the form $n = 2k$ where k is an integer. Therefore the sequence of even numbers $\{a_n\}$ can be represented with $a_n = 2n$ for all n .

Summations

A summation is the operation of adding a sequence of numbers. If numbers are added sequentially from left to right, any intermediate result is a partial sum, prefix sum or running total of the summation. The numbers to be summed called addends may be integers, rational numbers or real numbers.

For the summation of the sequence of consecutive integers from 1 to 100 one could use an addition expression involving an ellipsis to indicate missing terms: $1+2+3+4+5+\dots+99+100$. We use sigma operator called summation operator " Σ ".

Example: Express the sum of first 100 terms of even numbers sequence $a_n = 2n$ in terms of summation notations.

Solution: The lower limit for the index of summation is 1 and upper limit is 100.

Hence

$$\sum_{i=1}^{100} 2n$$

Series: The summation of an infinite sequence of values is called a series.

Some useful summation formulae

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{k=1}^n k(k+1) = \frac{n(n+1)(n+2)}{3}$$

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$$

$$\sum_{k=1}^n k(k+1)(k+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

$$\sum_{k=1}^n \frac{1}{k(k+1)(k+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

$$\sum_{k=1}^n (2k-1) = n^2$$

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Lab Work

Lab 1

Write a C program to find union of two sets.

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
    int A[] = {1,2,3,4,5,6,7,8};
    int B[] = {2,4,6,8,10,12};
    int C[14];
    int i,j,k=0,t;
    for(i=0;i<8;i++)
    {
        C[k++]=A[i];
```

```

}
for(i=0;i<6;i++)
{
    t=0;
    for(j=0;j<8;j++)
    {
        if(B[i]==A[j])
        {
            t=1;
            break;
        }
    }
    if(t==0)
    {
        C[k++]=B[i];
    }
}
printf("Set A = {");
for(i=0;i<8;i++)
{
    printf("%d",A[i]);
    if(i==7)
        continue;
    printf(",");
}
printf("}\n");
printf("Set B = {");
for(i=0;i<5;i++)
{
    printf("%d",B[i]);
    if(i==4)
        continue;
    printf(",");
}
printf("}\n");
printf("The Union set C = {");
for(i=0;i<k;i++)
{
    printf("%d",C[i]);
    if(i==k-1)
        continue;
    printf(",");
}

```



```

    }
    printf("}\n");
    getch();
    return 0;
}

```

Output:

Set A = {1,2,3,4,5,6,7,8}

Set B = {2,4,6,8,10}

The Union set C = {1,2,3,4,5,6,7,8,10,12}

Write a C program to find intersection of sets

Ans:

```

#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
    int A[] = {1,2,3,4,5,6,7,8};
    int B[] = {2,4,6,8,10};
    int C[5];
    int i,j,k=0;
    for(i=0;i<8;i++)
    {
        for(j=0;j<5;j++)
        {
            if(A[i]==B[j])
            {
                C[k++] = A[i];
            }
        }
    }
    printf("Set A = {");
    for(i=0;i<8;i++)
    {
        printf("%d",A[i]);
        if(i==7)
            continue;
        printf(",");
    }
    printf("}\n");
    printf("Set B = {");
    for(i=0;i<5;i++)
    {

```

```

    printf("%d",B[i]);
    if(i==4)
        continue;
    printf(",");
}
printf("\n");
printf("The intersection set C = {}");
for(i=0;i<k;i++)
{
    printf("%d",C[i]);
    if(i==k-1)
        continue;
    printf(",");
}
printf("\n");
getch();
return 0;
}

```

Output

Set A = {1,2,3,4,5,6,7,8}

Set B = {2,4,6,8,10}

The intersection set C = {2,4,6,8}

Write a C program to find difference of two sets

Ans:

```

#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
    int A[] = {1,2,3,4,5,6,7,8};
    int B[] = {2,4,6,8,10};
    int C[5];
    int i,j,k=0,t;
    for(i=0;i<8;i++)
    {
        t=0;
        for(j=0;j<5;j++)
        {
            if(A[i]==B[j])
            {
                t=1;
            }
        }
    }
}

```

```

    }
    if(t==0)
        C[k++]=A[i];
    }
    printf("Set A = {");
    for(i=0;i<8;i++)
    {
        printf("%d",A[i]);
        if(i==7)
            continue;
        printf(", ");
    }
    printf("}\n");
    printf("Set B = {");
    for(i=0;i<5;i++)
    {
        printf("%d",B[i]);
        if(i==4)
            continue;
        printf(", ");
    }
    printf("}\n");
    printf("The set C= A-B = {");
    for(i=0;i<k;i++)
    {
        printf("%d",C[i]);
        if(i==k-1)
            continue;
        printf(", ");
    }
    printf("}\n");
    getch();
    return 0;
}

```

Output:

Set A = {1,2,3,4,5,6,7,8}

Set B = {2,4,6,8,10}

The set C= A-B = {1,3,5,7}

Write a C program to find Cartesian product of two sets.

Ans:

```
#include<stdio.h>
```

```

#include<conio.h>
#include<math.h>
int main()
{
    int A[] = {1,2,3,4,5,6,7,8};
    int B[] = {2,4,6,8,10};
    int C[5];
    int i,j,k=0,t;
    printf("Set A = {");
    for(i=0;i<8;i++)
    {
        printf("%d",A[i]);
        if(i==7)
            continue;
        printf(",");
    }
    printf("}\n");
    printf("Set B = {");
    for(i=0;i<5;i++)
    {
        printf("%d",B[i]);
        if(i==4)
            continue;
        printf(",");
    }
    printf("}\n");
    printf("The cartesian product of A and B = {");
    for(i=0;i<8;i++)
    {
        for(j=0;j<5;j++)
        {
            printf("(%d,%d)",A[i],B[j]);
        }
    }
    printf("}\n");
    getch();
    return 0;
}

```

Output:

Set A = {1,2,3,4,5,6,7,8}

Set B = {2,4,6,8,10}

The Cartesian product of A and B = {(1,2),(1,4),(1,6),(1,8),(1,10),(2,2),(2,4),(2,6),(2,8),(2,10),(3,2),(3,4),(3,6),(3,8),(3,10),(4,2),(4,4),(4,6),(4,8),(4,10),(5,2),(5,4),(5,6),(5,8),(5,10),(6,2),(6,4),(6,6),(6,8),(6,10),(7,2),(7,4),(7,6),(7,8),(7,10),(8,2),(8,4),(8,6),(8,8),(8,10) }

Write a C program to find ceiling and floor value of a variable or expression

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
    float x = 2.456;
    float c = ceil(x);
    float f = floor(x);
    printf("The floor value of %f is = %f\n",x,f);
    printf("The ceiling value of %f is = %f\n",x,c);
    getch();
    return 0;
}
```

output:

The floor value of 2.456000 is = 2.000000

The ceiling value of 2.456000 is = 3.000000

Write a C program to find to implement fuzzy set operations

Ans:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
float min(float a,float b);
float max(float a,float b);
int main()
{
    float fa,fb,fi,fu,fac;
    float x = 1.0;
    printf("Enter membership function of first set:\n");
    scanf("%f",&fa);
    printf("Enter membership function of second set:\n");
    scanf("%f",&fb);
    fi = min(fa,fb);
    fu = max(fa,fb);
    fac = x-fa;
    printf("The membership function of intersection = %0.1f\n",fi);
}
```

```

printf("The membership function of union = %0.1f\n",fu);
printf("The membership function of complement of first set = %0.1f\n",fac);
getch();
return 0;
}
float min(float a, float b)
{
    if(a<b)
        return a;
    else
        return b;
}
float max(float a, float b)
{
    if(a>b)
        return a;
    else
        return b;
}

```

Output:

```

Enter membership function of first set:
0.2
Enter membership function of second set:
0.4
The membership function of intersection = 0.2
The membership function of union = 0.4
The membership function of complement of first set = 0.8

```

Program to find union and intersection of two sets using bit string representation

```

#include<stdio.h>
#include<conio.h>
#define N 10
int main()
{
    int i,k=0,j,t;
    int U[N]= {1,2,3,4,5,6,7,8,9,10};
    int A[N] = {1,3,4,5,7};
    int B[N] = {1,2,4,6,8,10};
    int SA[N];
    int SB[N];

```

```

int US[N];
int CS[N];
int DS[N];
int C[N];
int D[N];
for(i=0;i<N;i++)
{
    if(U[i]!=-1)
        US[i]=1;
}
printf("The Bit String representation of Universal set\n");
for(i=0;i<N;i++)
printf("%d ",US[i]);
printf("\n");
//Finding bit string for set A
for(i=0;i<N;i++)
{
    t=0;
    for(j=0;j<N;j++)
    {
        if(A[j]==U[i])
        {
            t=1;
            break;
        }
    }
    if(t==1)
        SA[i]=1;
    else
        SA[i]=0;
}
printf("The Bit String representation of set A\n");
for(i=0;i<N;i++)
printf("%d ",SA[i]);
printf("\n");

for(i=0;i<N;i++)
{

```

```

t=0;
for(j=0;j<N;j++)
{
    if(B[j]==U[i])
    {
        t=1;
        break;
    }
}
if(t==1)
SB[i]=1;
else
SB[i]=0;
}
//Finding bit string for set B
printf("The Bit String representation of set B\n");
for(i=0;i<N;i++)
printf("%d ",SB[i]);
printf("\n");
//computing union
for(i=0;i<10;i++)
{
    if(SA[i]==1 | SB[i]==1)
    CS[i]=1;
    else
    CS[i]=0;
}
for(i=0;i<10;i++)
{
    if(CS[i]==1)
    C[k++]=U[i];
}
//computing intersection
for(i=0;i<10;i++)
{
    if(SA[i]==1&&SB[i]==1)
    DS[i]=1;
    else

```



```

        DS[i]=0;
    }
    printf("The union set is\n");
    for(i=0;i<k;i++)
        printf("%d ",C[i]);
    k=0;
    for(i=0;i<10;i++)
    {
        if(DS[i]==1)
            D[k++]=U[i];
    }
    printf("\nThe intersection set is\n");
    for(i=0;i<k;i++)
        printf("%d ",D[i]);
    getch();
    return 0;
}

```

Output:

The Bit String representation of Universal set

1 1 1 1 1 1 1 1 1 1

The Bit String representation of set A

1 0 1 1 1 0 1 0 0 0

The Bit String representation of set B

1 1 0 1 0 1 0 1 0 1

The union set is

1 2 3 4 5 6 7 8 10

The intersection set is

1 4

Assignment 1

1. State and prove De Morgan's law in set theory

2. Show that if A and B are set then prove that $A-B = A \cap B'$
3. Draw Venn Diagrams for the following set: $(A \cap B) \cup (A \cap C)$
4. How sets are represented by using bit strings? Why it is preferred over unordered representation of sets.
5. Let $f(x) = ax+b$ and $g(x) = cx+d$ where a,b,c and d are constants. Determine for which constants a,b,c and d it is true that $f \circ g = g \circ f$
6. Show that a function $f(x) = ax+b$ from $\mathbb{R}$ to $\mathbb{R}$ is invertible where a and b are constants with $a \neq 0$ and find the inverse of f.
7. How can you relate domain and co-domain of functions with functions in programming language? Discuss composite and inverse of function with suitable function with examples
8. State inclusion-exclusion principle in set theory. A total of 1232 students have taken a course in Spanish, 879 have taken a course in French and 114 have taken course in Russian. Further, 103 have taken courses in both Spanish and French. 23 have taken courses in both Spanish and Russian and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French and Russian, how many students have taken a course in all three languages?
9. In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the number of families which buy
 - a. Number of families which buy all three newspapers.
 - b. Number of families which buy newspaper A only
 - c. Number of families which buy newspaper B only
 - d. Number of families which buy newspaper C only
 - e. Number of families which buy None of A, B, C
 - f. Number of families which buy exactly only one newspaper
 - g. Number of families which buy newspaper A and B only
 - h. Number of families which buy newspaper B and C only
 - i. Number of families which buy newspaper C and A only
10. Find a at least three different sequences beginning with the terms 3,5,7 whose terms are generated by a simple formula or rule.